UNIVERSITY OF CALGARY
FACULTY OF SCIENCE
Department of Physics and Astronomy

Uncertainty in the Pendulum

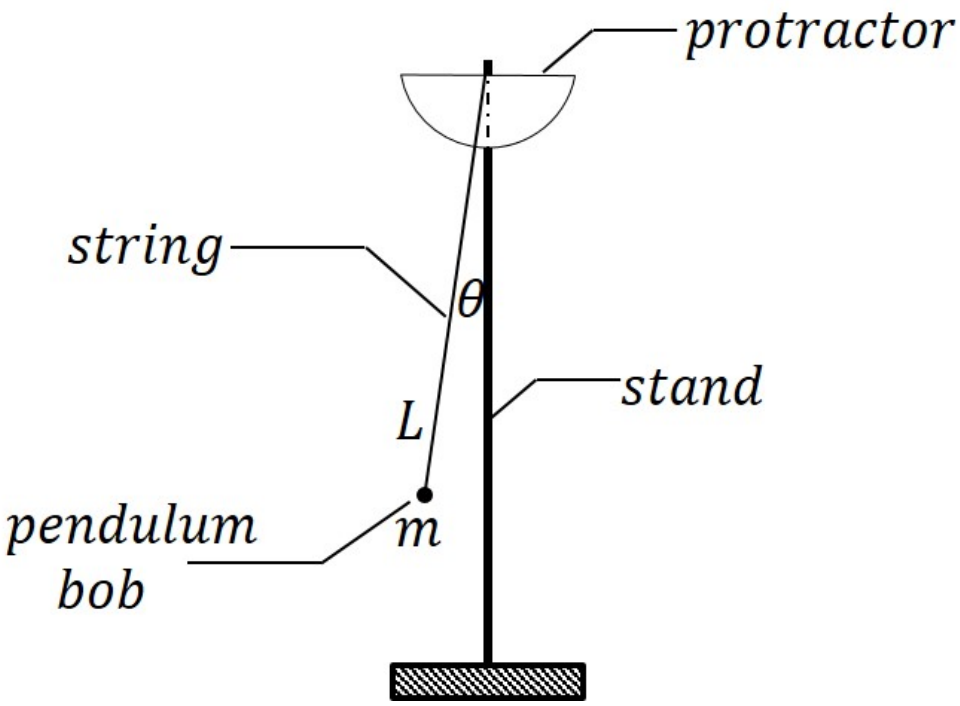


Figure 1.1. Pendulum setup.

Equipment: String, set of hanging masses, laboratory stand, stopwatch, protractor, meter stick, digital balance, aluminum bar [2], right angle clamp [3], fork clamp, c-clamp [2].

Goals of the Experiment: To define and determine type A and type B uncertainty for performed measurements, to determine the uncertainty of a calculated value using uncertainty propagation, to explore various methods of assessing the uncertainty, and to determine the acceleration due to gravity, g, and its uncertainty.

Theory

Background

A simple pendulum is a mass attached to a string. The time needed for one full oscillation (period, T) is related to the length of the pendulum L and acceleration due to gravity g. The relationship is given by the following equation:

$$T = 2\pi\sqrt{\frac{L}{g}}. \tag{1.1}$$

The pendulum can be used to determine the acceleration due to gravity by measuring its length and period of oscillation.

Uncertainty is associated with each measured quantity and it quantifies the dispersion of the values that could reasonably be attributed to the measured quantity. There are two types of uncertainties: type A (derived through repeated measurements) and type B (NOT derived through repeated measurements).

In each part of the experiment you will measure the length of the pendulum and its period (with uncertainties) in order to calculate the acceleration due to gravity and determine its uncertainty. The four values of g you will find will be compared to the expected value within uncertainties. The accuracy of each part of the experiment will be determined.

The Experiment

Pre-Lab

Your Pre-lab exercises are available on the D2L Lab site (Assessment – Quizzes) and should be completed there.

Preliminary Measurements

Human Reaction Time

In this section we will investigate the reaction time of you and your classmates. This is important to share the actual value as it is not a competition but an exploratory exercise that will allow you to properly estimate uncertainties in this experiment.
Each member of the group should complete the following task: start the stopwatch and attempt to stop it exactly at $t = 5.00$ s. Record the value of the difference between the reading and 5.00 s mark (e.g. 4.90 s - 5.00 s = -0.10 s) in the Excel template. Repeat three times for each group member. Pass the results of each group member to your TA. The results of your measurements will be displayed, providing the average reaction time of your lab section and the standard deviation associated with the distribution. This data will help you to properly determine the uncertainty in the time of the pendulum swing.

Oscillation time

[Discussion 1] Explain in your own words what an oscillation is. Discuss how the period of one oscillation is defined and record your discussion in the Excel template.

Length of the pendulum

In this section we will investigate the uncertainty in the length measurement of the pendulum. It is important to allow everyone to take a measurement of the pendulum and make the estimates by themselves.

[Discussion 2] Discuss what the length of pendulum L stands for. Decide how you determine the length of your pendulum. Discuss with your group mates which mass you are going to use and why.

Experimental Procedure

Set up the pendulum as shown in Figure 1.2.

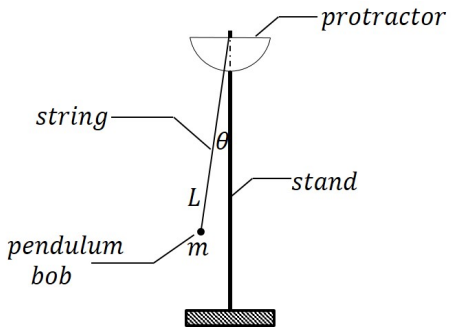


Figure 1.2. Experimental Setup.

Singular Measurement of an Oscillation Time

- Select a string of length of approximately 1.0 m. Attach the selected mass to the string. Measure the length L of the pendulum and record in the Excel template. Estimate an uncertainty in one measurement taking into account everything you discussed in Discussion 2.
- Measure the time T for **one oscillation** of the pendulum. Make sure to use a small angle (less than 10 degrees) for all oscillations in this lab. Record that value and determine its uncertainty $u(T)$ in the Excel template (use the information about the reaction time from the preliminary measurements).
- Using the oscillation time T and the length of the pendulum L, rearrange Equation 1.2 to calculate the acceleration due to gravity g_1. Use uncertainty propagation to calculate uncertainty $u(g_1)$.

Multiple Measurements of an Oscillation Time

- Repeat the measurement of a single oscillation ten times. Each group member should measure the time of one full swing of the pendulum at least twice (group of four) or three times (group of three). Record the values in the Excel template. Estimate the uncertainty of each time measurement.
- Determine the average value of the period of the pendulum $\bar{T}$ and its uncertainty.
  [Discussion 3] In this part of the lab, we are deriving a Type A uncertainty for the period. Does this encapsulate all of your uncertainty about the measurement? Discuss whether the uncertainty in the length would change if the length was measured 10 times. How would it affect the uncertainty calculation?
- Using the average period of the oscillation $\bar{T}$ and the length of the pendulum L determined in the previous section, calculate the acceleration due to gravity g_2 and its uncertainty $u(g_2)$.

Singular Measurement of Time of Multiple Oscillations

- Measure the time t of ten oscillations of the pendulum and estimate the uncertainty in your time measurement. Record your results in the Excel template.
- Using the time t of ten oscillations, calculate the period of one oscillation T with the uncertainty $u(T)$. Use the uncertainty propagation to calculate $u(T)$.
- Using the oscillation time T and the length of the pendulum L from the first section, determine the acceleration due to gravity g_3 with uncertainty $u(g_3)$.

Measurements of Multiple Oscillations for Various Pendulum Lengths

[Discussion 4] Discuss with your group mates six different values of length you are going to use and why.

- Set up a pendulum of a given length L. Record the length and its uncertainty.
- For each length, measure the time of ten full oscillations of the pendulum. Perform this experiment only once for each length. Record the values in the Excel template. Estimate the uncertainty of each time measurement.

Consider the Equation 1.2 relating the length of the pendulum to its period,

$$T = 2\pi\sqrt{\frac{L}{g}}. \tag{1.2}$$

This equation is not in the form $y = mx + b$, so a graph of T versus L will not be a straight line. However, we can easily rearrange the equation into a form that can be fit by a straight line. (See "Nonlinear relationships" in the "Measurement Uncertainty" document.)
[Discussion 5] Discuss with your group how Equation 1.2 could be rearranged so the plot of period (or function of it) versus length (or function of it) would be in the form $y = mx + b$. In the Excel template define specifically the expression for y, m, x and b. How will you find the value of g using the graph?

- Have your TA check your Discussion 5 notes.
- Using the Excel template create a graph of variables you have discussed (include error bars). Fit the straight line to the data. Determine and record the values of slope and y-intercept with uncertainties. Comment on the agreement of the y-intercept with the expected value.
- Determine the value of g_4 and its uncertainty $u(g_4)$ from the slope of the graph.

[Discussion 6] Discuss with your group mates what are the advantages and disadvantages of the graphical analysis.

Discussion Items

- Compare the values of acceleration due to gravity calculated in each part of the experiment within their uncertainties. Decide on the type of uncertainty in period, length, and gravitational acceleration. Present your results in a table.
- Natural Resources Canada measures gravitation due to gravity at various points. The station on the University of Calgary campus reports the value to be $g = 9.8080$ m/s^2 with uncertainty $u(g) = 0.0001$ m/s^2. Compare the calculated values to the accepted value. Comment on the agreement. List sources of errors specifying whether they are random or systematic.
- The combination of equatorial bulge (the difference between polar and equatorial diameters of Earth) as well as the effects of Earth's rotation and gravitational pull of the Moon and the Sun causes variation of the acceleration due to gravity between 9.780 $\frac{m}{s^2}$ at the Equator to 9.832 $\frac{m}{s^2}$ at the poles. Is the precision of this experiment sufficient to notice this difference?

Data Requirements

Before leaving the laboratory you have to submit the Excel template provided as a record of your measurements and discussions.

Lab Report

Your lab report should include the following items:

- Brief theoretical introduction quoting relationships between physical variables.
- Tables including all the measurements with units and uncertainties. You can use the tables from the Excel templates as guides for organizing your data.
- A graph showing the dependence of period on the length of the string with values of slope and y-intercept. Explain how you rearranged Equation 1.2 to be in the form $y = mx + b$.
- Calculations of four values of the acceleration due to gravity with uncertainties (one from each part of this experiment).
- Table including the type of uncertainty in the period and the pendulum length measurements and calculations in each part of this experiment.
- Answers/comments to all discussion items.
- Your conclusion about the value of the acceleration due to gravity, its comparison to the accepted value and any comments on agreement/disagreement.